

Citibank & Aereve: HKMA GenAI Sandbox Success

Validated in HKMA Sandbox | Supported by Cyberport AISC

Citibank: The Challenge & Why Sandbox

High Volume of False Positives: The increasing use of Chinese names in our processes, particularly due to character variations (e.g., 峰 / 峯), exacerbates the false positive problem

HKMA IPMN Scheme: Partnered with HK Baptist University and Aereve as dedicated mentor, nurturing FinTech talent through practical industry experience

Manual Review Inefficiency: Manual reviews of large volumes of unstructured data are time-consuming, error-prone, and resource-intensive

Cyberport AISC: Leveraged H800 GPU resources for model fine-tuning on 500 real-world AML cases

Hurdles Encountered

- **Information Security:** Sharing bank data is difficult; establishing secure server required significant investment
- **Data Labelling:** Extensive human intervention needed for quality training data

Solution & Results

Three Key AML/KYC Tasks:

1. **Relevance Classification:** Determine if matched information refers to the same entity
2. **Sentiment Analysis:** Classify sentiment as Positive/Neutral/Negative for risk assessment
3. **Text Summarization:** Extract key AML/KYC information and risk indicators

81.2%
Relevance

92.4%
Sentiment

69%
ROUGE-L

Performance Comparison

Baseline (Llama)	66.5%
DeepSeek-V3	75.7%
Fine-tuned (Ours)	81.2%

Citibank: Would We Do It Again? YES!

Invaluable learning experience. The 80/20 Rule: If we can potentially achieve an 80% reduction of time and resource, why not?

"If we start again, are we going to do it given all the hurdles? YES! Invaluable learning experience."

— Edgar Ma, Cluster AML Head of Greater China and Korea, Citibank

HKMA Gen A.I. Sandbox Final Report

*Enhancing On-Premise AML/KYC System with GenAI:
Improving Accuracy, Efficiency, and Compliance*

by Citi & Aereve 2025-08-18

3. Solution Overview

3.1 Data Collection and Preparation

- Data Source:** The dataset comprised the input customer profile and google search results of 500 AML screen cases. Each screen entity has 10 matched links, and the resulted data has approximately **5,000 Q&A pairs**. The information gathered for each pair of screen case include:
 - Input profile: entity name, type, country, gender, date of birth/registration date, identification number, profile notes, etc. (as show in Fig. 3);
 - Matched information: google link title, snippet, and link content (obtained via web crawler as shown in Fig. 4 and 5).

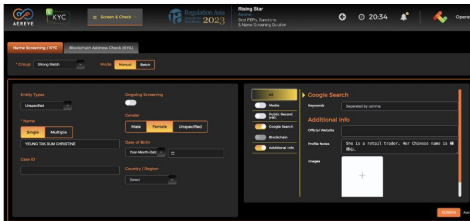


Fig. 3 Name screen page of E2KYC platform for the customer profile input.

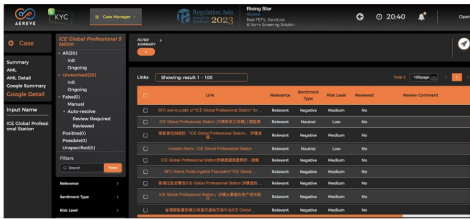


Fig. 4 An example of name screening web search result links overview.

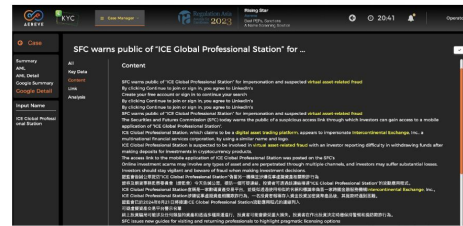


Fig. 5 An example of name screening web search link content details.

The questions of each pair of data include three important AML/KYC tasks including:

- Relevance Classification:** determine whether the matched information is referring to the same input entity (as shown in Fig. 6).
 - Relevant:** The matched profile or article is referring to the same entity.
 - Irrelevant:** The matched information indicating a different entity.
 - Not Sure:** Information is insufficient to make decisions on the relevance.
- Sentiment Classification:** determine the sentiment type of the matched information.
 - Positive:** Indicates favourable information about the matched entity (e.g., awards, achievements).
 - Neutral:** Contains factual or balanced information.
 - Negative:** Indicates adverse or AML risk-related information (e.g., sanction, lawsuits, fraud accusations).
- Text Summarization:**
 - Focus on extracting key information related to AML/KYC (e.g., risk indicators, person/entity mentions).

4. Experiments and Results

This section provides a detailed overview of the experiments conducted, the evaluation metrics used, and the results achieved during the fine-tuning of **Llama-3.2-3B-Instruct** for AML/KYC tasks. The results are compared against the baseline model and an industry-standard benchmark (**DeepSeek V3**) to demonstrate the effectiveness of the fine-tuned model. The evaluation focuses on relevance, coherence, and alignment with human-reviewed summaries.

4.1 Experimental Procedure

- Baseline Evaluation:**
 - The pre-trained **Llama-3.2-3B-Instruct** model (without fine-tuning) was evaluated as the baseline.
 - Results indicated limited performance on AML/KYC-specific tasks due to lack of domain-specific adaptation.
- Benchmark Testing:**
 - The **DeepSeek-V3** model, widely used in AML/KYC workflows, was tested on the same dataset for comparison.
- Fine-Tuning with LoRA:**
 - The **Llama-3.2-3B** model was fine-tuned using low-rank adaptation (LoRA) for 2 days on two Cyberport H800 GPUs.
 - The fine-tuned model was then tested on the validation and test datasets.
- Evaluation and Comparison:**
 - All models were evaluated on the same test dataset using the specified metrics.

4.2 Results

4.2.1 Overall Results

The fine-tuned **Llama-3.2-3B-Instruct** model consistently outperforms the baseline model across all tasks as shown in Table 1, confirming the value of domain-specific fine-tuning. For **relevance classification** and **sentiment analysis**, the fine-tuned model achieves state-of-the-art performance, surpassing DeepSeek-V3 by **5.5%** and **5.9%**, respectively. For **text summarization**, DeepSeek-V3 maintains the best due to the alignment of human-reviewed summaries with its outputs, but the fine-tuned model still demonstrates substantial improvement over the baseline. These findings demonstrate the fine-tuned model's potential as a competitive alternative to industry-standard solutions, particularly in tasks involving classification and sentiment analysis.

Table 1. Experimental results of models in specific AML/KYC tasks

- Negative sentiment precision (78.4%)** is the weakest metric, indicating the model sometimes misclassifies cases as negative, potentially due to overlaps with other sentiment categories.
- The **weighted average F1-score (89.7%)** and accuracy (**92.4%**) confirm robust and reliable sentiment analysis across all classes, making the fine-tuned model highly effective for this task.

Table 3. Sentiment Classification Evaluation Results of Fine-tuned Model

Metric/Class	Positive	Negative	Neutral	Weighted Average
Precision	84.9	78.4	97.4	86.9
Recall	94.6	93.2	91.6	93.1
F1-Score	89.5	85.2	94.4	89.7
Accuracy	-	-	-	92.4

Sentiment Classification Confusion Matrix (Percentages)

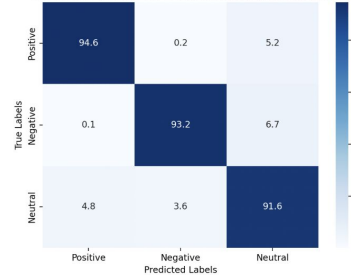


Fig.8 Sentiment classification confusion matrix of fine-tuned model.

4.2.3 Text Summarization

- Baseline:** Scored **32.1%** ROUGE-L, reflecting poor alignment with human-reviewed summaries due to its lack of domain knowledge.
- Benchmark:** Achieved the highest score of **68.4%**, benefiting from its foundational role in generating the initial summaries that were later reviewed and modified by human compliance officers.
- Fine-Tuned Model:** Scored **53.8%** ROUGE-L, with an significant improvement of **21.7%** compared to the original Llama-3.2-3B-Instruct (Baseline) indicating much better summary relevance and coherence.